Systems of linear equations

Linear Algebra

Licenciatura em Engenharia Informática

Instituto Superior de Engenharia do Porto

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Definition 1 (Linear Equation)

A **linear equation** in n unknows (or variables) is an equation that can be written in the form:

$$a_1x_1 + a_2x_2 + \cdots + a_{n-1}x_{n-1} + a_nx_n = b. \tag{1}$$

a_i, $i = 1, \ldots, n$, are real constants called **coefficients**, the real constant b is called **independent term** and x_i, $i = 1, \ldots, n$, are the **unknows** (or variables).
If we set $b = 0$ in equation (1) we obtain the equation

$$a_1x_1 + a_2x_2 + \cdots + a_{n-1}x_{n-1} + a_nx_n = 0, \tag{2}$$

called **homogeneous equation** associated to (1).



- *One solution of equation (1), is a n-tuple*

$$\boldsymbol{s} = (s_1, s_2, \ldots, s_{n-1}, s_n)$$

such that, if we substitute in (1), each variable x_i with s_i, $i = 1, \ldots, n$, we obtain a true statement.

- *If **s'** e **s"** are two solutions of equation (1) It follows that their difference $\boldsymbol{s} = \boldsymbol{s'} - \boldsymbol{s"}$ is a solution of the associated homogeneous equation(2) and every multiple of **s** is also a solution of the associated homogeneous equation (2).*

Examples:

- The equation $3x - y = 1$, in the unknowns x e y, is linear.
- Its associated homogeneous equation is the equation $3x - y = 0$.
- It follows that, $\mathbf{s'} = (1,2)$ e $\mathbf{s"} = (0,-1)$ are two solutions of the equation $3x - y = 1$.
- $\mathbf{s} = \mathbf{s'} - \mathbf{s"} = (1,3)$ is a solution of the associated homogeneous equation and for all $\lambda \in \mathbb{R}$, $\lambda\mathbf{s} = (\lambda, 3\lambda)$ is also a solution of the associated homogeneous equation.
- The equations $x - y^2 = 4$, $x + 2^y = 3$, $\cos(x) - y = 2$, e $xy + 2x = 1$ are not linear equations.

Definition 2 (System of linear equations (SLE))

A system of m linear equations in n unknowns is a set of m of the form

$$\{a_{i,1}x_1 + a_{i,2}x_2 + \cdots + a_{i,n-1}x_{n-1} + a_{i,n}x_n = b_i, \quad i = 1, \ldots, m\}$$

Usually we write

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n-1}x_{n-1} + a_{1,n}x_n &= b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n-1}x_{n-1} + a_{2,n}x_n &= b_2 \\ \vdots & \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n-1}x_{n-1} + a_{m,n}x_n &= b_m \end{cases} \tag{3}$$

If we set $b_i = 0$, $i = 1, \ldots, m$ in the system (3) we obtain their associated homogeneous system

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n-1}x_{n-1} + a_{1,n}x_n &= 0 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n-1}x_{n-1} + a_{2,n}x_n &= 0 \\ \vdots & \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n-1}x_{n-1} + a_{m,n}x_n &= 0 \end{cases} \tag{4}$$

☞ *Analogous to linear equations, we refer to the coefficients $a_{i,j}$, $i = 1 \ldots m$, $j = 1 \ldots n$ as the coefficients of the system of linear equations (SLE), the terms x_j, $j = 1 \ldots n$ as the unknowns or variables of the SLE, and the coefficients b_i, $i = 1 \ldots m$ as the independent terms.*

Definition 3 (Solution and solution set (of a SLE))

1. One **solution** s of the SLE (3) is a n-tuple $\mathrm{s} = (s_1, \ldots, s_n)$ that satisfy all the equations in the system (3).
2. The **Set Solution** (SS) of a SLE is the set of all possible solutions that satisfy the system of equations. In other words, if the system of linear equations (SLE) has solutions, it is a set of n-tuples, if the SLE has no solutions, the solution set is the empty set.

Examples:

1. The SLE of two equations and two unknowns, x and y

$$\begin{cases} x+y=1 \\ -x+2y=-1 \end{cases} \tag{5}$$

only has one solution, $s=(1,0)$. The we write $SS=\{(1,0)\}$ or $SS=\{(x,y)\in\mathbb{R}^2 \; x=1\wedge y=0\}$.

2. The SLE of two equations and two unknowns, x e y

$$\begin{cases} x+y=1 \\ 2x+2y=0 \end{cases} \tag{6}$$

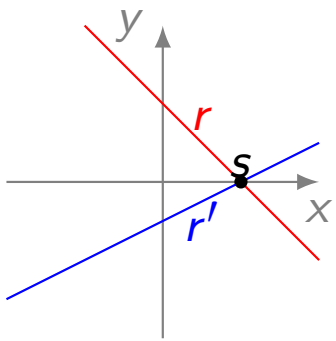
has no solutions.In this case we write, $SS=\emptyset$.

3. The SLE of two equations and two unknowns, x e y

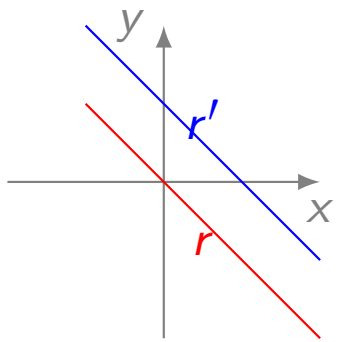
$$\begin{cases} x+y=1 \\ 2x+2y=2 \end{cases} \tag{7}$$

It has infinitely many solutions. In fact, for every $\alpha\in\mathbb{R}$ the ordered pair $(\alpha,1-\alpha)$ is a solution of (7). In this case, we write $SS=\{(x,y)\in\mathbb{R}^2 : y=1-x\}$ (here x is a free variable and y is a dependent variable). Alternatively, we can write $SS=\{(x,1-x)\in\mathbb{R}^2 \; x\in\mathbb{R}\}$.

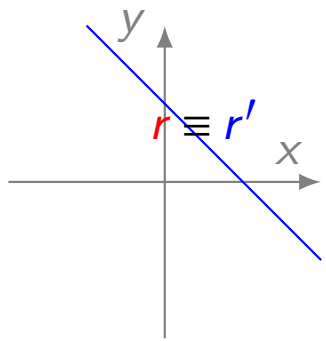
Geometrically, the solution sets of these systems of linear equations (SLE) are formed by the intersection points of two lines r and r' in R^2.



SLE(5), intersecting.



SLE (6), parallel.



SLE (7), coincident.

Property 1

The solution set of a SLE always has one of the following forms,

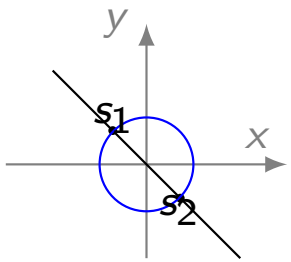
1. Is the empty set. In this case, we say that the SLE is *inconsistent* (IS).
2. It has only one solution. In this case, we say that the SLE is *consistent independent*(CIS).
3. It has infinite solutions. In this case, we say that the SLE is *consistent dependent*(CDS).

☞ *Note that a solution set of a system of linear equations cannot have a finite number of solutions greater than one.*

If we consider the system of 2 equations in the 2 variables x e y

$$\begin{cases} x+y=0 \\ x^2+y^2=1 \end{cases}$$

we find that geometrically the solution set is formed by the points of intersection, $s_1=(-\frac{\sqrt{2}}{2},\frac{\sqrt{2}}{2})$ and $s_2=(\frac{\sqrt{2}}{2},-\frac{\sqrt{2}}{2})$, of the line with the equation $y=-x$ and the circle centered at the origin with a unit radius.



This system of equations does not satisfy property 1 because the solution set has exactly two elements. This situation only occurs because this system of equations is not a system of linear equations (the second equation is not linear).

Definition 4 (Equivalent systems of linear equations)

Let S_1 and S_2 be two SLE (with the same number of linear equations and unknowns) with solution sets SS_1 and SS_2, respectively. We say that the systems S_1 e S_2 are *equivalents* iif

$$SS_1 = SS_2.$$

Consider the following SLE

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n-1}x_{n-1} + a_{1,n}x_n &= b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n-1}x_{n-1} + a_{2,n}x_n &= b_2 \\ \vdots & \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n-1}x_{n-1} + a_{m,n}x_n &= b_m \end{cases}. \tag{8}$$

Definition 5 (Matricial Form of a SLE)

The matricial form of the system of linear equations (SLE) (8) is defined as the matrix equation

$$\mathrm{A}\mathrm{x} = \mathrm{b},$$

onde

$$\mathrm{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n-1} & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n-1} & a_{2,n} \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n-1} & a_{m,n} \end{bmatrix}, \quad \mathrm{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathrm{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

- A is the **coefficient matrix**
- x is the **column matrix of variables**
- b is the **column matrix of constants**.

Definition 6 (Augmented Matrix of a SLE)

The **Augmented Matrix** of the SLE (8) is the coefficient matrix A augmented with the column matrix of constants b in the last column.

$$[\mathrm{A}|\mathrm{b}] = \left[\begin{array}{ccccc|c} a_{1,1} & a_{1,2} & \cdots & a_{1,n-1} & a_{1,n} & b_1 \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n-1} & a_{2,n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n-1} & a_{m,n} & b_m \end{array}\right]$$

☞ *The coefficient matrix of a SLE with m equations and n variables is a matrix with size $m \times n$ and it augmentable matrix has size $m \times (n+1)$.*

Example: The matricial form of the SLE

$$\begin{cases} 2x-3y+4t=2 \\ 6x-7z-t=5 \\ -3y+9t=\frac{5}{3} \end{cases}$$

is the matricial equation

$$\begin{bmatrix} 2 & -3 & 0 & 4 \\ 6 & 0 & -7 & -1 \\ 0 & -3 & 0 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ \frac{5}{3} \end{bmatrix}.$$

The augmented matrix is

$$[\mathrm{A}|\mathrm{b}] = \left[\begin{array}{cccc|c} 2 & -3 & 0 & 4 & 2 \\ 6 & 0 & -7 & -1 & 5 \\ 0 & -3 & 0 & 9 & \frac{5}{3} \end{array}\right].$$

Theorem 1 (classification of SLE)

Let $A \in M_{m,n}$ the coefficient matrix and $[A|b]$ be the augmented matrix of a SLE (with m equations and n unknowns. Then we have

$$\text{SEL:} \begin{cases} r(A) = r([A|b]) \implies \text{Consistent} \begin{cases} r(A) = r([A|b]) = n \implies \text{Independent} \\ r(A) = r([A|b]) < n \implies \text{dependent with degree } n - r(A) \end{cases} \\ r(A) < r([A|b]) \implies \text{Inconsistent} \end{cases}$$

☞ *Here, for simplicity, we write* r *instead of* rank.

☞ *The process of classifying a system of linear equations (SLE) outlined in Theorem 1 requires calculating the rank of the coefficient and augmented matrices of the SLE. This calculation can be performed using the Gaussian elimination algorithm. However, as we will see, this algorithm will also be useful for solving the SLE.*

Theorem 2

Let S be a SLE with augmented matrix [A|b]. If the matrix [A′|b′] results from [A|b] by applying a finite sequence of elementary row operations, then the SLE S' with augmented matrix [A′|b′] is equivalent to S.

Nota: In other words, the systems of linear equations S and S' have the same solution set.

Corolário 1

Let S be a SLE with augmented matrix [A|b] and $E([A|b])$ a row-equivalent row echelon form matrix to [A|b]. Then the SLE S' associated th the augmented matrix $E([A|b])$ is equivalent to the SLE S.



The solution of a system of linear equations (SLE), S' , whose augmented matrix is in row echelon form can be easily solved using the back substitution algorithm. Consequently, by solving S', we are solving S , because Theorem 1 guarantees that they have the same solution set. The following examples describe the solution of systems of linear equations via Gaussian elimination.

Example I (Consistent independent system): Considere o SEL

$$\begin{cases} x + 2y + 2z = 2 \\ 3x - 2y - z = 5 \\ 2x - 5y + 3z = -4 \\ x + 3y + 4z = 1 \end{cases}.$$

Começamos por aplicar o algoritmo de eliminação Gaussiana à matriz completa

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 3 & -2 & -1 & 5 \\ 2 & -5 & 3 & -4 \\ 1 & 3 & 4 & 1 \end{array}\right] \xrightarrow[L_4 \leftarrow L_4 - L_1]{\substack{L_2 \leftarrow L_2 - 3L_1 \\ L_3 \leftarrow L_3 - 2L_1}} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & -8 & -7 & -1 \\ 0 & -9 & -1 & -8 \\ 0 & 1 & 2 & -1 \end{array}\right] \xrightarrow[\substack{\text{(swap only} \\ \text{to simplify} \\ \text{the calculations)}}]{L_2 \leftrightarrow L_4} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & -9 & -1 & -8 \\ 0 & -8 & -7 & -1 \end{array}\right] \rightarrow$$

$$\xrightarrow[L_4 \leftarrow L_4 + 8L_2]{L_3 \leftarrow L_3 + 9L_2} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 17 & -17 \\ 0 & 0 & 9 & -9 \end{array}\right] \xrightarrow{L_4 \leftarrow L_4 - \frac{9}{17} L_3} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 17 & -17 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Analyzing the last matrix (which is in row echelon form), we have

$$\mathrm{r}\left(\underbrace{\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 17 \\ 0 & 0 & 0 \end{bmatrix}}_{A'}\right) = \mathrm{r}\left(\underbrace{\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 17 & -17 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{[A'|b']}\right) = \text{number of unknowns } (3)$$

Hence, the system is **consistent independent**. To solve the system, we find the system of linear equations associated with the augmented matrix $[A'|b']$ by ignoring the null row.[1]

$$\begin{cases} x + 2y + 2z = 2 \\ \quad\; y + 2z = -1 \\ \quad\quad\; 17z = -17 \end{cases}$$

We substitute the value of z found from the third equation into the second equation and find the value of $z = -1$. We substitute the value of z found from the third equation into the second equation and find the value of $y = 1$. Finally, we find the value of $x = 2$, by substituting the unknowns y e z with the values obtained in the solution of the last equations into the first equation. And the set solution is

$$SS = \{(2, 1, -1)\}.$$

[1] the null row corresponds to the equation $0 = 0$, which is a universal condition.

Example II (Consistent dependent system):
Consider the following SLE with 5 equations and 4 unknowns x, y, z e t

$$\begin{cases} x + y + 2z + t = 1 \\ x + z + 2t = 3 \\ -y - z + t = 2 \\ 2x + 2z + 4t = 6 \\ y + t = 2 \end{cases}.$$

Applying Gaussian elimination to the augmented matrix [A|b] in order to obtain an equivalent matrix in row echelon form [A′|b′], we have

$$\left[\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 1 & 0 & 1 & 2 & 3 \\ 0 & -1 & -1 & 1 & 2 \\ 2 & 0 & 2 & 4 & 6 \\ 0 & 1 & 0 & 1 & 2 \end{array}\right] \xrightarrow[L_4 \leftarrow L_4 - 2L_1]{L_2 \leftarrow L_2 - L_1} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & -1 & -1 & 1 & 2 \\ 0 & -1 & -1 & 1 & 2 \\ 0 & -2 & -2 & 2 & 4 \\ 0 & 1 & 0 & 1 & 2 \end{array}\right] \xrightarrow[L_4 \leftarrow L_4 - 2L_2]{L_3 \leftarrow L_3 - L_2} \xrightarrow[L_5 \leftarrow L_5 + L_2]{}$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & -1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & 4 \end{array}\right] \xrightarrow{L_3 \leftrightarrow L_5} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 1 \\ 0 & -1 & -1 & 1 & 2 \\ 0 & 0 & -1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

- Hence, we have $\mathrm{r}(\mathrm{A}') = \mathrm{r}([\mathrm{A}'|\mathrm{b}']) = 3 < n(=4)$. Consequently, the system is **consistent dependent**.
- The **degree** is $n - \mathrm{r}(\mathrm{A}') = 1$. In this way, we will have **one free variable** in the description of the solution set.



The degree of a consistent dependent linear system is typically described in terms of the number of free variables present in the solution set.

The SLE associated with the augmented matrix $[\mathrm{A}'|\mathrm{b}']$, ignoring the null rows, is

$$\begin{cases} x + y + 2z + t = 1 \\ -y - z + t = 2 \\ -z + 2t = 4 \end{cases}.$$

1. The back substitution process begins by solving the last row in **order to the variable associated with the leading coefficient** (variable z).
2. In this way, we have$z = 2t - 4$ and the variable t is free.
3. Substituting the variable z with $2t-4$ in the second equation, we conclude that $y = -t+2$.
4. The variable x is obtained from the first equation, where we substitute the variable z with $2t-4$ and the variable y with $-t+2$. Performing the calculations, we obtain $x = -4t+7$.



The set solution is:

$$SS = \left\{(x,y,z,t) \in \mathbb{R}^4 : x = -4t+7 \wedge y = -t+2 \wedge z = 2t-4\right\}$$

*or in **parametric form***

$$SS = \left\{(-4t+7, -t+2, 2t-4, t) \in \mathbb{R}^4 : t \in \mathbb{R}\right\}.$$

Example III (Inconsistent system): Consider the LSE with 3 equations and 3 unknowns x, y e z

$$\begin{cases} & & y & - & z & = & 1 \\ 2x & + & y & - & z & = & 0 \\ 2x & + & 3y & - & 3z & = & 1 \end{cases}.$$

$$\left[\begin{array}{ccc|c} 0 & 1 & -1 & 1 \\ 2 & 1 & -1 & 0 \\ 2 & 3 & -3 & 1 \end{array}\right] \xrightarrow{L_1 \leftrightarrow L_2} \left[\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & 1 & -1 & 1 \\ 2 & 3 & -3 & 1 \end{array}\right] \xrightarrow{L_3 \leftarrow L_3 - L_1} \left[\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 2 & -2 & 1 \end{array}\right] \rightarrow$$

$$\xrightarrow{L_3 \leftarrow L_3 - 2L_2} \left[\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & -2 \end{array}\right].$$

As we have

$$2 = \mathrm{r}(\mathrm{A}') < \mathrm{r}([\mathrm{A}'|\mathrm{b}']) = 3$$

Then the SLE **is inconsistent** and the set solution is

$$SS = \emptyset.$$

☞ *Note that the last line gives rise to the equation* $0x + 0y + 0z = -2$, *which has no solution.*

Example IV (Classification of SLE as a function of parameters):
Consider the following system of 3 linear equations in the 3 incógnitas x, y e z, which depend on the real parameters a and b,

$$\begin{cases} x + y + z = 1 \\ x + 2y + az = b \\ 3y - z = b-1 \end{cases}.$$

Applying Gaussian elimination to the augmented matrix of the system of linear equations, we have

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & a & b \\ 0 & 3 & -1 & b-1 \end{array}\right] \xrightarrow{L_2 \leftarrow L_2 - L_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & a-1 & b-1 \\ 0 & 3 & -1 & b-1 \end{array}\right] \xrightarrow{L_3 \leftarrow L_3 - 3L_2} \underbrace{\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & a-1 & b-1 \\ 0 & 0 & -3a+2 & -2(b-1) \end{array}\right]}_{[\mathrm{A}'|\mathrm{b}']}$$

The following steps consist of:

(1) To verify under what conditions, regarding the parameters a and b, the rank of the coefficient matrix A' **takes the maximum value** (3 in this example).

- We have $\mathrm{rank}(\mathrm{A}') = 3$ iff $-3a+2 \neq 0$. That is the rank of A' takes its maximum value iff $a \neq \frac{2}{3}$.
- Under the condition $a \neq \frac{2}{3}$ it follows that the rank of the matrix $[\mathrm{A}'|\mathrm{b}']$ is also equal to 3, regardless of the value of the parameter b.

Hence we have

$$\mathrm{r}(\mathrm{A}') = \mathrm{r}([\mathrm{A}'|\mathrm{b}']) = \text{number of unknowns}(=3); \qquad a \neq \frac{2}{3} \wedge \forall\, b \in \mathbb{R}$$

(2) To determine under what conditions, regarding the parameters a e b, the rank of coefficient matrix A' is not maximum.

- It is sufficient to negate the condition for which the rank of A' is maximum. Thus, we have that the rank of A' is less than 3 iff $a = \frac{2}{3}$. And we have, $r(A') = 2$, regardless of the value of the parameter b.
- Imposing this condition on the matrix $[A'|b']$, we have

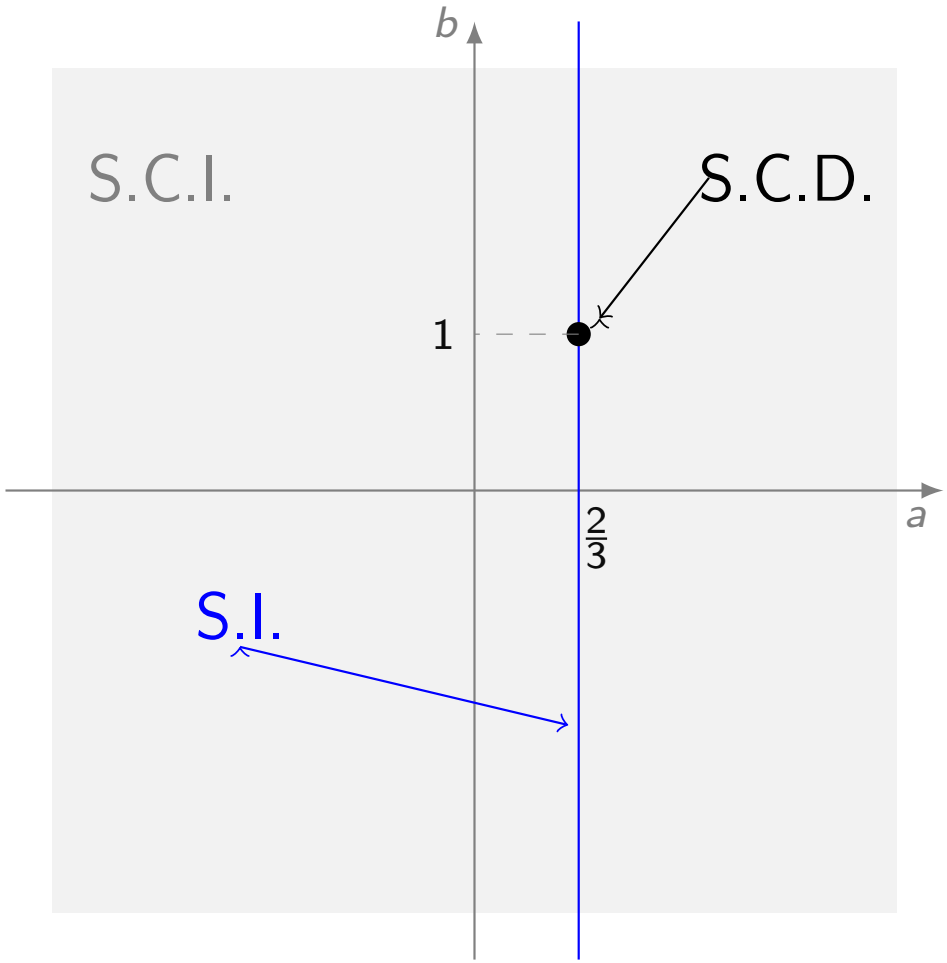
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -\frac{1}{3} & b-1 \\ 0 & 0 & 0 & -2(b-1) \end{array}\right]$$

We still need to verify what is the rank of the augmented matrix is, which depends on the value of the paramete b. It is easy to verify that $\text{rank}([A'|b'])$, can take two values

i) If $-2(b-1) \neq 0$ (or $b \neq 1$), we have $r([A'|b']) = 3$. Hence the SLE is inconsistent.

ii) If $-2(b-1) = 0$ (or $b = 1$), we have $r([A'|b']) = 2$. Hence the SLE is consistent dependent with degree equal to one.

We summarize this discussion in the following figure



The system is consistent independent in the entire plane except for the line $a = 2/3$, marked in the figure in blue color. In line $a = 2/3$ the system is inconsistent except at the point with coordinates $a = 2/3$ e $b = 1$ where the system is consistent dependent.

Definition 7 (Matrix in reduced row echelon form)

Consider the matrix $A \in \mathscr{M}_{m,n}$. We say that A is in **reduced row echelon form** if it satisfies the properties:

1. A is in row echelon form;
2. All leading coefficients of A are unitary;
3. If a column of A has a leading coefficient, then the leading coefficient is the only non-zero entry in that column.

Examples: The following matrices are all in reduced row echelon form

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{e} \quad \begin{bmatrix} 1 & 2 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Algorithm 1 (Gauss-Jordan Elimination)

Consider the following matrix $A \in \mathscr{M}_{m,n}$.

1. Apply the Gaussian elimination algorithm. A matrix E in row echelon form is obtained.
2. We normalize all the leading coefficients. This forces the leading coefficient c_ℓ of all row ℓ to be unitary, multiplying this row by $\frac{1}{c_\ell}$.
3. We eliminate all the entries $e_{i,j}$ that are above each leading coefficient c_ℓ by performing the elementary operation $L_i \leftarrow L_i - e_{i,j} L_\ell$.

Example: To solve the following system of linear equations with 4 and 5 unknowns x_i, $i = 1, 2, \ldots, 5$ using Gauss-Jordan elimination.

$$\begin{cases} 2x_1 + 2x_2 - 3x_3 + x_5 = 0 \\ 4x_1 + 2x_2 - 6x_3 - x_4 - 2x_5 = 2 \\ 6x_1 + 4x_2 - 9x_3 - x_4 - x_5 = 2 \\ 2x_2 + 2x_4 + x_5 = -1 \end{cases}.$$

1st **STEP**: Apply the Gaussian algorithm to the augmented matrix

$$\left[\begin{array}{ccccc|c} 2 & 2 & -3 & 0 & 1 & 0 \\ 4 & 2 & -6 & -1 & -2 & 2 \\ 6 & 4 & -9 & -1 & -1 & 2 \\ 0 & 2 & 0 & 2 & 1 & -1 \end{array}\right] \xrightarrow[L_3 \leftarrow L_3 - 3L_1]{L_2 \leftarrow L_2 - 2L_1} \left[\begin{array}{ccccc|c} 2 & 2 & -3 & 0 & 1 & 0 \\ 0 & -2 & 0 & -1 & -4 & 2 \\ 0 & -2 & 0 & -1 & -4 & 2 \\ 0 & 2 & 0 & 2 & 1 & -1 \end{array}\right] \xrightarrow[L_4 \leftarrow L_4 + L_2]{L_3 \leftarrow L_3 - L_2}$$

$$\rightarrow \left[\begin{array}{ccccc|c} 2 & 2 & -3 & 0 & 1 & 0 \\ 0 & -2 & 0 & -1 & -4 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -3 & 1 \end{array}\right] \xrightarrow{L_3 \leftrightarrow L_4} \begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \end{array} \\ \left[\begin{array}{ccccc|c} 2 & 2 & -3 & 0 & 1 & 0 \\ 0 & -2 & 0 & -1 & -4 & 2 \\ 0 & 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \end{array} \xrightarrow{2nd\ step}$$

- It follows that the SLE is consistent dependent, with degree 2

$$\text{rank}(A') = \text{rank}([A'|b']) = 3 < n = 5$$

- Consequently, there will be two free variables and three dependent variables in the solution set (SS).
- We can choose the variables , x_1,x_2 e x_4, as **dependent variables** , (**The variables associated with the columns that contain leading coefficients** marked with blue color) and the remaining variables x_3 e x_5 are **free**.

2nd Step: Normalize the leading coefficients (make the leading coefficients unitary).

$$\xrightarrow[L_2 \leftarrow -\frac{1}{2}L_2]{L_1 \leftarrow \frac{1}{2}L_1} \left[\begin{array}{ccccc|c} 1 & 1 & -\frac{3}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & \frac{1}{2} & 2 & -1 \\ 0 & 0 & 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{3rd \text{ Step } \ldots}$$

3rd Step: For each leading coefficient, eliminate the entries that share the same column.

$$\xrightarrow{L_2 \leftarrow L_2 - \frac{1}{2}L_3} \left[\begin{array}{ccccc|c} 1 & 1 & -\frac{3}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 0 & \frac{7}{2} & -\frac{3}{2} \\ 0 & 0 & 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{L_1 \leftarrow L_1 - L_2} \begin{array}{c} \\ \\ \\ \\ \end{array} \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \end{array} \left[\begin{array}{ccccc|c} 1 & 0 & -\frac{3}{2} & 0 & -3 & \frac{3}{2} \\ 0 & 1 & 0 & 0 & \frac{7}{2} & -\frac{3}{2} \\ 0 & 0 & 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Once the matrix is found in reduced row echelon form, we can immediately determine the solution set. WE have,

$$\text{Da } 3^{\text{a}} \text{ linha resulta,} \quad x_4 = 1 + 3x_5.$$

$$\text{Da } 2^{\text{a}} \text{ linha resulta,} \quad x_2 = -\frac{3}{2} - \frac{7x_5}{2}.$$

$$\text{Da } 1^{\text{a}} \text{ linha resulta,} \quad x_1 = \frac{3}{2} + \frac{3x_3}{2} + 3x_5$$

And the set solution of the SLE takes the form,

$$SS = \left\{ \left(\frac{3}{2} + \frac{3x_3}{2} + 3x_5, -\frac{3}{2} - \frac{7x_5}{2}, x_3, 1 + 3x_5, x_5 \right) \in \mathbb{R}^5 : x_3, x_5 \in \mathbb{R} \right\},$$

or alternatively.

$$SS = \left\{ (x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 : x_1 = \frac{3}{2} + \frac{3x_3}{2} + 3x_5 \wedge x_2 = -\frac{3}{2} - \frac{7x_5}{2} \wedge x_4 = 1 + 3x_5 \right\}.$$

Algorithm 2 (Calculation of the inverse matrix using Gauss-Jordan elimination)

1. Given an invertible matrix $A \in \mathscr{M}_n$ we construct the augmented matrix

$$[A|I_n] \in \mathscr{M}_{n,2n}$$

2. We apply the Gauss-Jordan algorithm to the matrix constructed above, thus obtaining its reduced row echelon form

$$\left[I_n|A^{-1}\right].$$

The inverse matrix of A is the block on the right side.

Example: Calculate the inverse of the matrix A.

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 3 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

We have,

$$\left[\begin{array}{cccc|cccc} 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 3 & 0 & 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{L_1 \leftrightarrow L_2} \left[\begin{array}{cccc|cccc} 2 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 3 & 0 & 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{L_3 \leftarrow L_3 - \frac{3}{2} L_1}$$

$$\left[\begin{array}{cccc|cccc} 2 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -3 & 3 & 0 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[L_3 \leftarrow -\frac{1}{3} L_3]{L_1 \leftarrow \frac{1}{2} L_1} \left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & \frac{1}{2} & -\frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{L_3 \leftarrow L_3 + L_4}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{3} & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{L_1 \leftarrow L_1 - L_3} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & -1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{3} & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right].$$

hence

$$A^{-1} = \begin{bmatrix} 0 & 0 & \frac{1}{3} & -1 \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{3} & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Definition 8 (Cramer's System)

A system of linear equations is called a *Cramer's system* if and only if

1. The number of equations is equal to the number of unknowns.
2. The determinant of the coefficient matrix is non-zero.

☞ *The translation to English is: "In a Cramer system, the system's matrix is always a square matrix. Therefore, it makes sense to consider the determinant of the system's matrix.*

Property 2

A Cramer's system is always consistent independent.

Methods for solving Cramer's systems:

(1) Writing a Cramer system in its matrix form $Ax = b$ tit follows that the system's matrix is invertible. Thus, the solution is given by

$$x = A^{-1} b.$$

☞ *This method should only be applied in theoretical problems.*

(2) The second method for solving Cramer systems is to use the **Cramer's formulas**. Writing a Cramer system in its matrix form $Ax = b$, where

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

It follows that the value of the k-th unknown x_k, $1 \leq k \leq n$, is given by

$$x_k = \frac{\det(\boldsymbol{A'_k})}{\det(A)},$$

where the matrix $\boldsymbol{A'_k}$ results from substituting the k-th colunm of the coefficient matrix A with colunm b of the independent coefficients.

Example: We aim to solve the following Cramer system

$$\begin{cases} x_1 + x_2 + x_3 = 3 \\ x_1 - x_2 = 5 \\ x_1 + x_3 = 9 \end{cases}.$$

The coefficient matrix A and the matrix b of independent coefficients matrix are

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{e} \quad b = \begin{bmatrix} 3 \\ 5 \\ 9 \end{bmatrix}$$

Aso $\det(A) = -1 \neq 0$ It follows that the system is a Cramer's system, and the values of the unknowns are given by

$$x_1 = \frac{\det(\boldsymbol{A}'_1)}{\det(A)} = \frac{\begin{vmatrix} 3 & 1 & 1 \\ 5 & -1 & 0 \\ 9 & 0 & 1 \end{vmatrix}}{-1} = \frac{1}{-1} = -1$$

$$x_2 = \frac{\det(\boldsymbol{A}'_2)}{\det(A)} = \frac{\begin{vmatrix} 1 & 3 & 1 \\ 1 & 5 & 0 \\ 1 & 9 & 1 \end{vmatrix}}{-1} = \frac{6}{-1} = -6$$

$$x_3 = \frac{\det(\boldsymbol{A}'_3)}{\det(A)} = \frac{\begin{vmatrix} 1 & 1 & 3 \\ 1 & -1 & 5 \\ 1 & 0 & 9 \end{vmatrix}}{-1} = \frac{-10}{-1} = 10.$$